Revision – Exponential Functions, Finance and Trig intro

Grade 10 **August 2018**

WORK TO TWO DECIMAL PLACES WHERE NEEDED

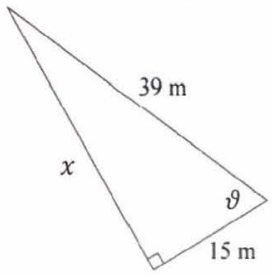
FORMULAE YOU MAY NEED:

$S.I = Pin$ $A = P(1 + i \times n)$ $A = P(1+i)^n$

$A = (1 - i \times n)$ $A = P(1-i)^n$

QUESTION 1

Use the following diagram to answer the questions which follow



a) Find the value of x, giving reasons (2)

$x^2 = 39^2 - 15^2$ (Pythag)

$x = 36$

b) Complete, leaving your answers in fractional form: **(3)**

$Sin\,\vartheta = \frac{36}{39} = \frac{12}{13}$

$Cos\,\vartheta = \frac{15}{39} = \frac{5}{13}$

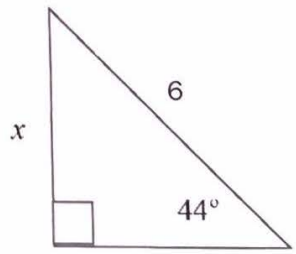
$Tan\,\vartheta = \frac{36}{15} = \frac{12}{5}$

[5]

QUESTION 2

Solve for the unknown in the following:

a) **(3)**

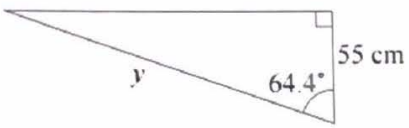


$\sin 44° = \frac{x}{6}$

$x = 6 \sin 44°$

$= 4,17$

b) **(3)**



$\cos 64,4° = \frac{55}{y}$

$y = \frac{55}{\cos 64,4°} = 127,29.$

[6]

QUESTION 3

a) If $\hat{A} = 155°$ and $\hat{B} = 324°$, use a calculator to determine the value of $\frac{\cos A}{2} + \sin 2B$ to two decimal places **(1)**

$-1,40$

b) $Tan^2\hat{A} =$ **(1)**

$0,22$

c) $\sin\frac{B}{2} =$ **(1)**

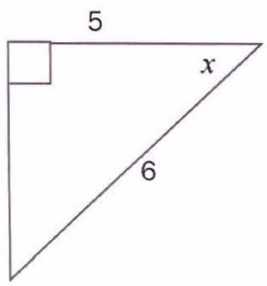
$0,31$

[3]

QUESTION 4

Solve for the unknown in the following:

a)

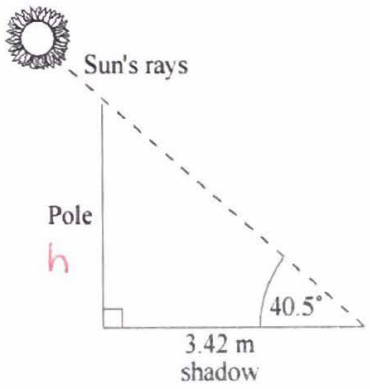


$\cos x = \frac{5}{6}$

$x = 33,56°$

(3)

b) The length of the shadow of a vertical pole is 3,42m long when the rays of the sun are inclined at an angle of $40,5^{\circ}$ to the horizontal. What is the height of the pole? Give the answer correct to two decimal places. **(3)**



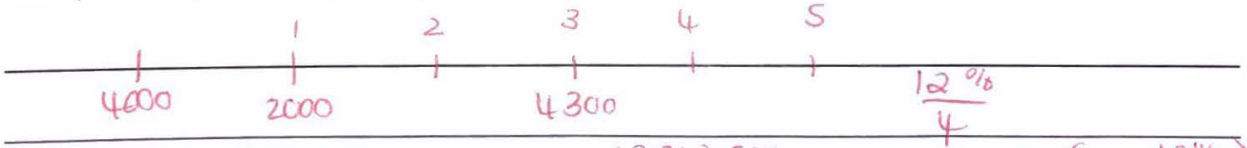
$\tan 40,5° = \frac{h}{3,42}$

$h = 2,92\ m.$

[6]

QUESTION 8

Lungelo decides to deposit any money she earns over the next five years into a savings account. She opens an account and deposits R4 000 immediately. One year later she deposits another R2 000 into the account . Two years later she deposits a further R4300 into the account.
Calculate the total amount of money she has saved at the end of 5 years if the interest rate was 12% p.a. compounded quarterly.



$A = P(1+i)^n \quad A = 4000\left(1+\frac{12\%}{4}\right)^{5\times4} + 2000\left(1+\frac{12\%}{4}\right)^{4\times4} + 4300\left(1+\frac{12\%}{4}\right)^{2\times4}$

$= R15880,97$

(6)